

Vasicek vs CR+

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1 Introduction

The Vasicek model and CreditRisk+ (CR+) are two of the most widely used frameworks for modelling credit portfolio risk. Although their mathematical formulations differ substantially, both models are designed to describe the same economic phenomenon: the impact of systematic credit risk on a large portfolio of obligors. In both frameworks, defaults become correlated because borrowers are exposed to a common risk driver representing the state of the economy. In the asymptotic limit of a large homogeneous portfolio, idiosyncratic risk is diversified away and portfolio losses are entirely determined by this common factor. Consequently, both models belong to the class of Asymptotic Single Risk Factor (ASRF) models. The principal difference between the models lies in how the systematic risk factor is represented:

- In the Vasicek model, systematic risk is described by a latent normally distributed factor Z and an asset correlation parameter ρ .
- In CreditRisk+, systematic risk is represented by a stochastic default intensity Λ , typically assumed to follow a gamma distribution.

Despite these differing representations, the models can be calibrated so that their loss distributions become nearly identical in the large portfolio limit. This can be achieved either by matching the variance of the common risk driver or by matching a specific quantile of the loss distribution, such as the 99% Value-at-Risk. Both approaches are discussed below.

2 Systematic Risk in Vasicek and CreditRisk+

Both models generate default dependence through a portfolio-wide systematic state variable. The specific mathematical constructions differ, but the economic interpretation is the same: favourable systematic states produce few defaults, while adverse states produce large clusters of defaults.

2.1 Vasicek Model

In the Vasicek model, the credit quality of borrower i is represented by the latent variable

$$X_i = \sqrt{\rho} Z + \sqrt{1 - \rho} \varepsilon_i,$$

where

$$Z \sim N(0, 1), \quad \varepsilon_i \sim N(0, 1),$$

and the idiosyncratic variables ε_i are mutually independent and independent of Z . A default occurs when

$$X_i < \Phi^{-1}(p),$$

where p denotes the unconditional probability of default. Conditioning on a realization $Z = z$ gives

$$P(X_i < \Phi^{-1}(p) \mid Z = z) = P\left(\sqrt{\rho}z + \sqrt{1-\rho}\varepsilon_i < \Phi^{-1}(p)\right).$$

Rearranging yields

$$P\left(\varepsilon_i < \frac{\Phi^{-1}(p) - \sqrt{\rho}z}{\sqrt{1-\rho}}\right).$$

Since ε_i is standard normally distributed,

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}z}{\sqrt{1-\rho}}\right).$$

This conditional probability of default is the key quantity governing portfolio losses in the asymptotic limit. A large positive value of z corresponds to a strong economy and a low conditional default rate, whereas a large negative value of z produces elevated default probabilities and clustered losses.

2.2 CreditRisk+

CreditRisk+ introduces systematic dependence through a stochastic default intensity. Conditional on an intensity realization $\Lambda = \lambda$, defaults are assumed to occur according to a Poisson process,

$$N \mid \Lambda = \lambda \sim \text{Poisson}(\lambda).$$

Thus,

$$E[N \mid \Lambda = \lambda] = \lambda, \quad \text{Var}(N \mid \Lambda = \lambda) = \lambda.$$

Systematic risk is introduced by allowing the intensity itself to vary between economic states. In the standard CreditRisk+ formulation,

$$\Lambda \sim \text{Gamma}(\alpha, \beta),$$

with shape parameter α and scale parameter β . The moments of the gamma distribution are

$$E[\Lambda] = \alpha\beta,$$

and

$$\text{Var}(\Lambda) = \alpha\beta^2.$$

The mean determines the unconditional expected number of defaults, while the variance controls the degree of systematic uncertainty around that expectation. A useful measure of systematic risk is the relative variance

$$CV^2 = \frac{\text{Var}(\Lambda)}{E[\Lambda]^2}.$$

Substituting the moments of the gamma distribution gives

$$CV^2 = \frac{\alpha\beta^2}{(\alpha\beta)^2},$$

which simplifies to

$$CV^2 = \frac{1}{\alpha}.$$

Hence, the shape parameter completely determines the relative variability of the common intensity factor. The economic interpretation is analogous to the Vasicek model. Realizations of Λ correspond to different systematic states:

- Low values of Λ correspond to favourable conditions with few defaults.
- Values near $E[\Lambda]$ correspond to average economic conditions.
- High values of Λ correspond to stressed conditions with large numbers of defaults.

Because all obligors share the same realization of Λ , defaults become correlated even though they are conditionally independent given the intensity.

2.3 Common Interpretation

The two models can therefore be viewed as alternative representations of the same underlying mechanism. In the Vasicek model, the common state variable is the Gaussian factor Z , which determines the conditional probability of default

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho} Z}{\sqrt{1 - \rho}}\right).$$

In CreditRisk+, the common state variable is the gamma-distributed intensity

$$\Lambda,$$

which directly determines the conditional default frequency. In the large portfolio limit, portfolio losses are fully determined by the realization of these common factors. The asymptotic loss rate may therefore be written schematically as

$$L_\infty = p(Z)$$

in the Vasicek model and

$$L_\infty \propto \Lambda$$

in CreditRisk+. To illustrate the similarity, consider a portfolio of 10 000 identical loans with an unconditional probability of default of 1%. The expected number of defaults is therefore 100 per year. In CreditRisk+, different realizations of the common intensity correspond to different economic states:

- A favourable state with a low realization of Λ , producing substantially fewer than 100 defaults.
- An average state with $\Lambda \approx E[\Lambda]$, producing approximately 100 defaults.
- An adverse state with a high realization of Λ , producing a large cluster of defaults.

Exactly the same interpretation can be made in the Vasicek model. Once the systematic factor takes a realized value $Z = z$, all borrowers share the same conditional probability of default

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho} z}{\sqrt{1 - \rho}}\right).$$

The realization of z therefore determines the overall state of the credit portfolio:

- A favourable state corresponds to a high value of z , for which $p(z)$ becomes substantially lower than 1% and only a small number of loans default.
- An average state corresponds approximately to $z \approx 0$, for which the default rate is close to its unconditional level and the number of defaults is around 100.
- An adverse state corresponds to a low value of z , for which $p(z)$ increases significantly and a large cluster of defaults occurs.

In both models, defaults are conditionally independent once the common risk factor has been fixed. In CreditRisk+, this factor is the stochastic intensity Λ , while in the Vasicek model it is the Gaussian factor Z . Default clustering therefore does not arise from direct interactions between obligors, but from their shared exposure to the same systematic state of the economy. From this perspective, the principal difference between the models is not the economic mechanism but the mathematical representation of the systematic risk driver. Vasicek models the state of the economy through a normally distributed latent factor, whereas CreditRisk+ models it through a gamma-distributed default intensity. Both approaches generate a single-factor loss distribution whose shape is determined by the variability of the common systematic risk factor.

3 Derivation of the Relationship between ρ and Gamma Variance

The discussion above showed that both Vasicek and CreditRisk+ represent systematic credit risk through a single common risk factor. In the Vasicek model this factor affects the portfolio through the conditional probability of default $p(Z)$, whereas in CreditRisk+ it acts through the stochastic default intensity Λ . In the large portfolio limit, portfolio losses are completely determined by these common risk drivers. A natural way to compare the two models is therefore to compare the variability of $p(Z)$ and Λ . By matching their relative variances, an approximate relationship between the Vasicek asset correlation ρ and the CreditRisk+ gamma variance can be derived.

3.1 Conditional PD as a random variable in the Vasicek model

In the Vasicek model, the conditional probability of default is given by

$$p(Z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho} Z}{\sqrt{1 - \rho}}\right),$$

where $Z \sim \mathcal{N}(0, 1)$ is the systematic factor. Since Z is random, $p(Z)$ is also a random variable. For low unconditional PD and moderate variations in Z , the function $p(Z)$ can be linearized around $Z = 0$:

$$p(Z) \approx p(0) + p'(0) Z.$$

The derivative is given by

$$p'(Z) = -\frac{\sqrt{\rho}}{\sqrt{1-\rho}} \phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho} Z}{\sqrt{1-\rho}}\right),$$

where ϕ is the density function of the standard normal distribution.

Evaluated at $Z = 0$, we obtain

$$p'(0) = -\frac{\sqrt{\rho}}{\sqrt{1-\rho}} \phi\left(\frac{\Phi^{-1}(p)}{\sqrt{1-\rho}}\right).$$

3.2 Approximation for low PD

For low values of p , the approximation holds¹

$$\phi\left(\frac{\Phi^{-1}(p)}{\sqrt{1-\rho}}\right) \approx \frac{1}{\sqrt{1-\rho}} \phi(\Phi^{-1}(p)).$$

Inserted into the expression above gives

$$p'(0) \approx -\sqrt{\rho} \phi(\Phi^{-1}(p)).$$

The variance of the conditional PD can thus be approximated as

$$\text{Var}[p(Z)] \approx (p'(0))^2 \text{Var}(Z) = \rho \phi(\Phi^{-1}(p))^2.$$

Dividing this by the square of the unconditional PD gives the relative variance:

$$\frac{\text{Var}[p(Z)]}{E[p(Z)]^2} \approx \rho \left(\frac{\phi(\Phi^{-1}(p))}{p}\right)^2.$$

For low PD, the ratio $\phi(\Phi^{-1}(p))/p$ grows at the same order as $(1-\rho)^{-1/2}$, which leads to the ASRF-established result

$$\frac{\text{Var}[p(Z)]}{E[p(Z)]^2} \approx \frac{\rho}{1-\rho}.$$

3.3 Relative variance in CreditRisk+

In CreditRisk+, the default intensity Λ is assumed to be gamma distributed. For a gamma distribution, it holds exactly that

$$\frac{\text{Var}(\Lambda)}{E[\Lambda]^2} = CV^2,$$

where CV^2 is the squared coefficient of variation, also called the gamma volatility.

¹In Vasicek and ASRF-based portfolio models, terms of the type $\phi(c/\sqrt{1-\rho})$ appear, where $c = \Phi^{-1}(p)$ is the default threshold and ρ denotes the degree of systematic dependence. For low probabilities of default, $c \rightarrow -\infty$, which implies that asymptotic tail approximations for the normal distribution can be applied. In this regime, the approximation

$$\phi\left(\frac{c}{\sqrt{1-\rho}}\right) \approx \frac{1}{\sqrt{1-\rho}} \phi(c)$$

is often used. It is not exact but valid asymptotically for low default levels.

3.4 Identification in the large portfolio limit

In the large portfolio limit, the loss distribution is entirely driven by variation in the common risk driver. For the Vasicek model and CreditRisk+ to produce the same loss distribution, the relative variance of this risk driver must therefore be the same.

By identifying the stochastic PD in the Vasicek model with the intensity in CreditRisk+, we obtain

$$CV^2 \approx \frac{\rho}{1 - \rho}.$$

Solving with respect to ρ yields the commonly used relationship

$$\rho \approx \frac{CV^2}{1 + CV^2}.$$

This shows that the gamma volatility in CreditRisk+ is an (approximately) alternative parameterization of the same systematic credit risk as the asset correlation in the Vasicek model.

4 Calibrations

In this section, we consider two common calibration approaches linking the Vasicek model and CreditRisk+. The first approach matches the variance of the common risk driver, while the second matches a tail quantile such as the 99% Value-at-Risk.

4.1 Calibration via gamma variance and asset correlation

Assume that the asset correlation in the Vasicek model is $\rho = 10\%$. At a 99% confidence level, the outcome corresponds to a strongly negative realization of the systematic factor Z . The conditional PD then becomes significantly higher than 1%, which leads to a large cluster of defaults.

By exploiting the relationship derived earlier, we translate this into CreditRisk+ using

$$CV^2 \approx \frac{\rho}{1 - \rho}.$$

In the numerical example,

$$\rho = 0.10.$$

Inserted into the expression above yields

$$CV^2 \approx \frac{0.10}{1 - 0.10} = \frac{0.10}{0.90} \approx 0.111.$$

By definition, the coefficient of variation satisfies

$$CV = \frac{\sqrt{\text{Var}(\Lambda)}}{E[\Lambda]}.$$

Thus,

$$CV = \sqrt{CV^2} = \sqrt{0.111} \approx 0.333.$$

Economic interpretation This implies that the standard deviation of the common default intensity Λ is approximately 33% of its mean. At the 99% quantile, a high value of Λ is realized, leading to essentially the same number of defaults as in the Vasicek model. Numerical comparisons typically show that the 99% VaR differs only marginally between the two models.

4.2 Calibration of ρ to a Given VaR Level

In practical applications, it may be desirable not only to match the variance of the common risk driver but also to match a specific quantile of the loss distribution. This is particularly relevant when CreditRisk+ is used as the primary model and one wishes to determine the corresponding asset correlation in the Vasicek framework. Assume that CreditRisk+ produces a 99% loss level (VaR) of

$$L_{0.99}^{CR+} = \ell_{99}.$$

We seek the value of ρ such that the Vasicek model produces the same portfolio loss at the 99% confidence level:

$$L_{0.99}^{Vasicek}(\rho) = \ell_{99}.$$

Portfolio loss in the Vasicek model For a homogeneous portfolio with identical exposures and constant loss given default (LGD), the asymptotic portfolio loss conditional on the systematic factor realization $Z = z$ is

$$L(z) = \text{LGD} \cdot p(z),$$

where

$$p(z) = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}z}{\sqrt{1-\rho}}\right)$$

is the conditional probability of default. To simplify the notation, we set

$$\text{LGD} = 1$$

in the following. The derivation is unchanged for any other constant LGD, since the loss distribution is then obtained by a simple rescaling. Under this normalization,

$$L(z) = p(z).$$

The 99% loss level Since portfolio losses are a decreasing function of the systematic factor Z , the 99% loss quantile corresponds to the 1% lower-tail realization of Z ,

$$z_{0.01} = \Phi^{-1}(0.01).$$

The Vasicek 99%-VaR is therefore

$$L_{0.99}^{Vasicek} = p(z_{0.01}),$$

which yields

$$L_{0.99}^{Vasicek} = \Phi\left(\frac{\Phi^{-1}(p) - \sqrt{\rho}\Phi^{-1}(0.01)}{\sqrt{1-\rho}}\right).$$

Using the symmetry property

$$\Phi^{-1}(0.01) = -\Phi^{-1}(0.99),$$

we obtain the familiar ASRF representation

$$L_{0.99}^{Vasicek} = \Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho}\Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right).$$

Solution for ρ To match the CreditRisk+ VaR, we solve

$$\Phi\left(\frac{\Phi^{-1}(p) + \sqrt{\rho} \Phi^{-1}(0.99)}{\sqrt{1-\rho}}\right) = \ell_{99}.$$

Applying the inverse standard normal distribution function to both sides gives

$$\frac{\Phi^{-1}(p) + \sqrt{\rho} \Phi^{-1}(0.99)}{\sqrt{1-\rho}} = \Phi^{-1}(\ell_{99}).$$

Rearranging,

$$\sqrt{\rho} \Phi^{-1}(0.99) = \Phi^{-1}(\ell_{99})\sqrt{1-\rho} - \Phi^{-1}(p).$$

This equation can be solved analytically or numerically for ρ . Since the Vasicek VaR is a monotonic increasing function of the asset correlation, the solution is unique. For a constant LGD different from unity, the calibration condition becomes

$$\text{LGD} \cdot L_{0.99}^{\text{Vasicek}} = \ell_{99},$$

or equivalently

$$L_{0.99}^{\text{Vasicek}} = \frac{\ell_{99}}{\text{LGD}}.$$

Thus, a constant LGD merely rescales the loss distribution and does not affect the underlying relationship between VaR and the asset correlation.

5 Conclusion

The Vasicek model and CreditRisk+ are two representations of the same underlying systematic risk. With appropriate calibration, they produce nearly identical loss distributions in large portfolios, especially at high confidence levels such as 99%.